Software Requirements Specification: Hospital Management System

PES University, Bangalore | Department of Computer Science and Engineering

Version 1.0 | Group 11 | Date created: 02-10-2026

Prepared by: Navya Suresh (PES1UG24CS904), Daksha S G (PES1UG24CS137), Dhruv Swatantramath (PES1UG24CS155), Gnanasagar H (PES1UG24CS171)

Revision History

Name	Date	Reason for changes	Version
Navya Suresh (PES1UG24CS904)	02-10-2026	Initial draft for Mini-Project Part-1	1.0

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) describes the requirements of the Hospital Management System (HMS), release 1.0. HMS is a patient record and appointment management system for hospitals. This SRS covers the whole product as planned for the first release: authentication and access control, patient registration and records, appointment scheduling, clinical (consultation and prescription) records, billing, reports and administration.

1.2 Scope

Intended Audience

This document is intended for the project team (developers), testers who derive the Test Plan from it, the designers of the Software Architecture and Design Specification, the course instructors who evaluate the project, and hospital stakeholders who confirm that the requirements match their needs. Section 1 introduces the product, Section 2 gives the overall description, and Section 3 lists the detailed, testable requirements.

Product Scope

HMS replaces paper registers and spreadsheets in a small or medium hospital. Its objectives are (a) to keep one reliable electronic record per patient, (b) to remove double-booked and lost appointments, (c) to give doctors quick access to a patient's history and allergies, (d) to produce accurate bills and receipts, and (e) to protect patient data. The following are out of scope for release 1.0: inpatient bed management, pharmacy stock, laboratory and radiology systems, insurance claim processing, payroll, and medical-device integration.

1.3 Definitions, acronyms, and abbreviations

Term	Meaning
HMS	Hospital Management System
SRS	Software Requirements Specification
PHI	Protected Health Information: any data that identifies a patient together with health information
EMR	Electronic Medical Record
RBAC	Role-Based Access Control
TLS	Transport Layer Security
API	Application Programming Interface
UPI	Unified Payments Interface (Indian instant payment system)
RTO	Recovery Time Objective

1.4 References

- IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications (and ISO/IEC/IEEE 29148:2018).
- PES University SRS template and Project Guidelines, Dept. of CSE, 2026.
- Problem Statement 11: Hospital Management System, Software Engineering Mini-Project Deliverables Part-1, ETA 07-10-2026.
- Digital Personal Data Protection Act, 2023 (India).
- OWASP Application Security Verification Standard (ASVS) 4.0.
- Companion documents: Software Test Plan (HMS) v1.0 and Software Architecture and Design Specification (HMS) v1.0.

1.5 Overview

Section 2 describes the product in context: its perspective, major functions, users, constraints and assumptions. Section 3 gives the external interface requirements, the functional requirements organised by system feature (with the UML use case diagram), the non-functional requirements (performance, availability, usability, security and others) and design constraints. Each requirement has a unique ID (HMS-F-nnn functional, HMS-NF-nnn non-functional, HMS-SEC-nnn security) so that it can be traced to design and to test cases (Appendix C).

2. Overall Description

2.1 Product perspective

HMS is a new, self-contained product. It is a client-server system: a desktop client (staff console and patient portal) communicates over HTTPS with a server application written in

C++ that stores data in a relational database. It does not depend on any existing hospital system in release 1.0, but its design keeps a notification adapter (SMS or e-mail) as an optional external interface.

2.2 Product functions

- Authenticate users and enforce role-based access (Section 3.2.2, FG1).
- Register patients and maintain their demographic records (FG2).
- Book, reschedule and cancel doctor appointments without double-booking (FG3).
- Record consultations, diagnoses and prescriptions, with drug-allergy warnings, and show patient history to doctors and to the patient concerned (FG4).
- Generate bills, record payments and produce receipts (FG5).
- Generate appointment and revenue reports, manage users and view the audit log (FG6).

2.3 User characteristics

User class	**Characteristics**	**Main functions used**	**Importance**
Receptionist	Daily use, moderate computer skills, needs fast screens	Register patients, search, book/cancel appointments	High
Doctor	Daily use, little time per patient, medical expertise but not IT	Schedule, record visits, prescribe, view history	High
Patient	Occasional use, varied skills, may use only the portal	Book/cancel own appointments, view own records and bills	Medium
Billing Officer	Daily use, accounting background	Bills, payments, receipts, revenue report	Medium
Administrator	Technical staff, privileged access	User management, audit log, reports	Medium

2.4 Constraints

Operating environment

Server: Ubuntu 22.04 LTS, 4 CPU cores, 8 GB RAM, PostgreSQL 15. Client: Windows 10/11 or Ubuntu 22.04 desktop with a display of at least 1366 x 768. Hospital local network with 100 Mbps or faster links.

Design and implementation constraints

- The problem statement requires implementation in C or C++; HMS shall use C++17.
- Patient data is personal data and shall be handled according to the DPDP Act 2023 and hospital policy.
- All communication shall use TLS 1.2 or higher.
- The team has about three months and no budget for commercial licences, so open-source libraries shall be used.

2.5 Assumptions and dependencies

- Doctors' working hours and consultation fees are entered by the Administrator before go-live.
- The hospital provides a reliable local network and a server machine with a UPS.
- Drug names and allergy names are entered as free text with a pick-list; no external drug database is used in release 1.0.
- Payment is recorded manually by the Billing Officer after the payment is received; HMS does not connect to a payment gateway.
- Open-source libraries (Qt 6, libpqxx, OpenSSL, an HTTP server library) remain available under their current licences.

3. Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces

- A desktop GUI with a consistent layout: menu on the left, work area on the right, status bar showing user and role.
- Every screen has a Help button and a Logout button; Enter and Tab keys move between fields; Ctrl+F opens patient search.
- Errors are shown in a red message bar next to the field concerned, in plain language, with no technical codes.
- Mandatory fields are marked with an asterisk; font size is at least 11 pt and colour contrast meets WCAG 2.1 AA.

3.1.2 Software Interfaces

Interface	**Purpose**	**Data in / out**
PostgreSQL 15 (via libpqxx)	Persistent storage of all data	SQL queries; patient, appointment, visit, bill, user and audit records
OpenSSL 3	TLS and AES-256 encryption	Keys, ciphertext

Interface	Purpose	Data in / out
Operating system file system	Backups and exported reports	Encrypted backup files, PDF and CSV files
Notification adapter (optional)	Appointment reminders by SMS or e-mail	Appointment ID, time, recipient contact

3.1.3 Communications Interfaces

Client and server communicate with HTTPS (TLS 1.2+) using a REST API and JSON messages. The server listens on one configurable port. Session tokens are sent in an Authorization header.

3.1.4 Hardware Interfaces

The only hardware interface is a standard printer used for receipts and reports. No medical devices are connected in release 1.0.

3.2 Functional Requirements

3.2.1 Analysis Model: UML Use Case Diagram

Figure 1 is the UML use case diagram of HMS. The actors Receptionist, Patient, Doctor, Billing Officer and Administrator are connected to use cases UC-01 to UC-12. Every use case requires the actor to log in first (HMS-F-001), so each use case includes the Login function. The summary table below lists each use case with its primary actor and related requirements.

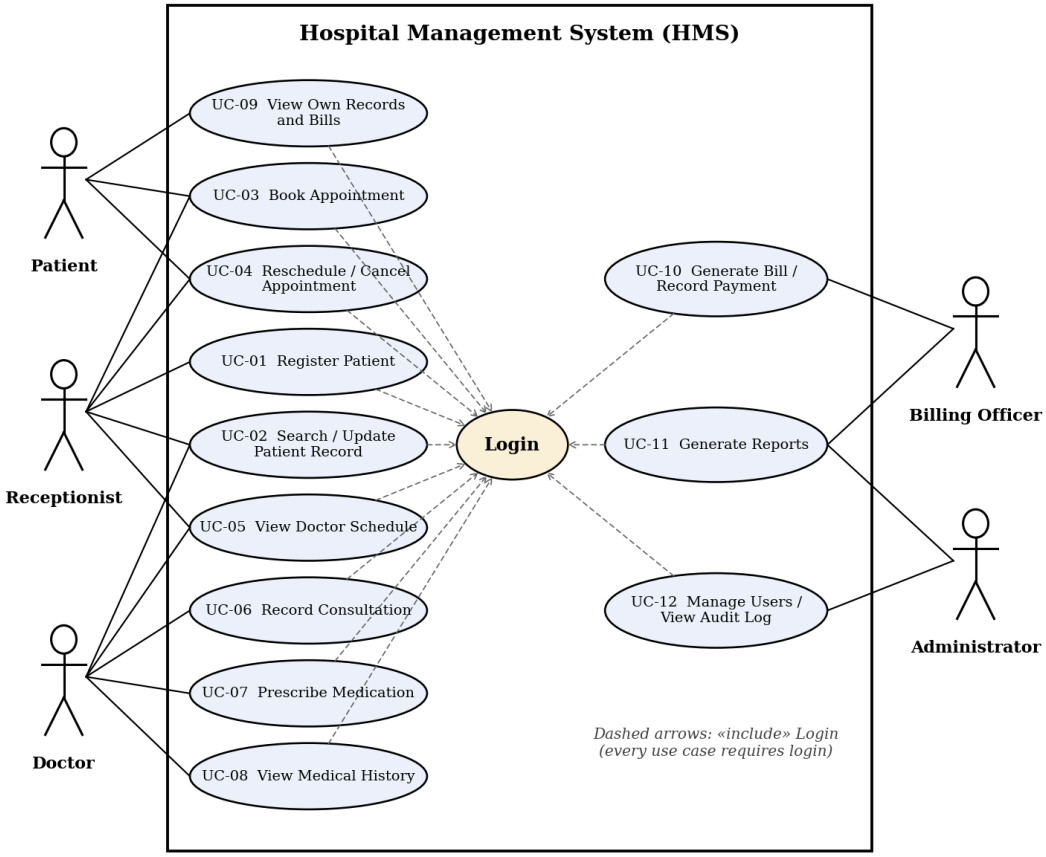


Figure 1. UML use case diagram of HMS. Dashed arrows show that every use case includes Login.

Use case summary

ID	Use case	Primary actor	Related requirements
UC-01	Register Patient	Receptionist	HMS-F-010, F-011, F-012
UC-02	Search / Update Patient Record	Receptionist, Doctor	HMS-F-013, F-014
UC-03	Book Appointment	Patient, Receptionist	HMS-F-020, F-021, F-022
UC-04	Reschedule / Cancel Appointment	Patient, Receptionist	HMS-F-023
UC-05	View Doctor Schedule	Doctor, Receptionist	HMS-F-024
UC-06	Record Consultation	Doctor	HMS-F-030

ID	Use case	Primary actor	Related requirements
UC-07	Prescribe Medication	Doctor	HMS-F-031, F-032
UC-08	View Medical History	Doctor	HMS-F-033
UC-09	View Own Records and Bills	Patient	HMS-F-003, F-034
UC-10	Generate Bill / Record Payment	Billing Officer	HMS-F-040, F-041, F-042
UC-11	Generate Reports	Administrator, Billing Officer	HMS-F-050
UC-12	Manage Users / View Audit Log	Administrator	HMS-F-060, F-061, F-003

3.2.2 System Features

Requirements are written as "The system shall …" statements, each with one measurable condition so that it can be verified by a test (see the Test Plan). Priority is High, Medium or Low.

FG1. Authentication and Access Control

Description and priority: Controls who can enter the system and what each role may do. Priority: High.

Stimulus / response sequence: User enters credentials; system validates them, creates a session and opens the dashboard of the user's role; on failure it shows an error and counts the attempt.

Req. ID	Requirement	Priority
HMS-F-001	The system shall authenticate every user with a unique username and password before granting access.	High
HMS-F-002	The system shall lock a user account for 15 minutes after 5 consecutive failed login attempts.	High
HMS-F-003	The system shall enforce role-based	High

Req. ID	Requirement	Priority
	access so that each role (Patient, Receptionist, Doctor, Billing Officer, Administrator) can use only the functions marked as permitted for it in the role-permission matrix below; any other request shall return "Access denied".	
HMS-F-004	The system shall respond to invalid credentials with the message "Invalid username or password" without revealing which field was wrong.	High

Role-permission matrix (HMS-F-003)

Y = permitted; Own = permitted for the user's own records only; - = not permitted (request returns "Access denied"). Each row can be tested by calling the function as every role.

Function	Patient	Receptionist	Doctor	Billing Officer	Administrator
Log in and log out	Y	Y	Y	Y	Y
Register patient (F-010 to F-012)	-	Y	-	-	-
Search patients (F-013)	-	Y	Y	-	-
Update patient data (F-014)	-	Y	-	-	-
View free slots (F-020)	Y	Y	-	-	-
Book appointment (F-020 to F-022)	Own	Y	-	-	-
Reschedule or cancel (F-023)	Own	Y	-	-	-
View own schedule (F-024)	-	-	Y	-	-
Record visit and prescribe (F-030 to F-032)	-	-	Y	-	-
View patient history (F-033)	-	-	Y	-	-
View own history and bills (F-034)	Own	-	-	-	-
Generate bill, record payment, receipt (F-040 to F-042)	-	-	-	Y	-
Generate reports (F-050)	-	-	-	Y	Y
Create and manage users (F-060)	-	-	-	-	Y
View audit log (F-061)	-	-	-	-	Y

FG2. Patient Management

Description and priority: Creates and maintains the patient record that all other features use. Priority: High.

Stimulus / response sequence: Receptionist fills the registration form; system validates, checks duplicates, saves and shows the new Patient ID. Search: user types a query; system lists matches.

Req. ID	**Requirement**	**Priority**
HMS-F-010	The system shall register a new patient by capturing name, date of birth, gender, phone number and address, and shall generate a unique Patient ID of the form PID-nnnnnnnn.	High
HMS-F-011	The system shall reject registration, and name the offending field, when a mandatory field is empty, the phone number is not exactly 10 digits, or the date of birth is in the future.	High
HMS-F-012	The system shall warn the user before saving when an existing patient has the same name, date of birth and phone number.	Medium
HMS-F-013	The system shall search patients by Patient ID, full or partial name, or phone number and list all matches, or show “No patient found”.	High
HMS-F-014	The system shall allow authorised staff to update patient demographic data and	Medium

Req. ID	Requirement	Priority
	shall retain every previous value with the timestamp and user who changed it.	

FG3. Appointment Management

Description and priority: Lets patients and staff reserve doctor time slots and lets doctors see their day. Priority: High.

Stimulus / response sequence: User selects doctor and date; system shows free slots; user selects a slot; system confirms and returns an Appointment ID, or refuses with the reason.

Req. ID	Requirement	Priority
HMS-F-020	The system shall let a user book an appointment by selecting a doctor, a date and an available 30-minute slot, and shall return a unique Appointment ID with status “Confirmed”.	High
HMS-F-021	The system shall prevent two confirmed appointments for the same doctor and slot, and shall display the next 3 available slots when a requested slot is taken.	High
HMS-F-022	The system shall reject bookings for a past date or time and for dates more than 60 days ahead.	Medium
HMS-F-023	The system shall allow an appointment to be rescheduled or cancelled up to 2 hours before its start time and shall reject such changes after that point.	Medium

Req. ID	Requirement	Priority
HMS-F-024	The system shall show a doctor the list of own appointments for a selected day, sorted by start time.	Medium

FG4. Clinical Records

Description and priority: Stores each consultation and prescription and shows history to doctors and, for their own records, to patients. Priority: High.

Stimulus / response sequence: Doctor opens a confirmed appointment, enters visit data and prescriptions; system checks allergies, saves, and shows the updated history.

Req. ID	Requirement	Priority
HMS-F-030	The system shall let a doctor record, for a visit linked to an appointment, the symptoms, diagnosis and notes, stamped with date, time and doctor ID.	High
HMS-F-031	The system shall let a doctor add prescription items (medicine, dosage, frequency, duration in days) to a visit.	High
HMS-F-032	The system shall display a warning when a prescribed medicine matches a recorded allergy of the patient and shall save the item only after the doctor enters an override reason.	High
HMS-F-033	The system shall display a patient's visits, diagnoses and prescriptions in reverse chronological	High

Req. ID	Requirement	Priority
	order to authorised doctors.	
HMS-F-034	The system shall let a logged-in Patient view only own appointments, visits, diagnoses, prescriptions and bills, newest first, and shall return "Access denied" for any other patient's record.	Medium

FG5. Billing

Description and priority: Turns a visit into a bill, records payments and produces receipts. Priority: High.

Stimulus / response sequence: Billing Officer opens a visit; system builds the bill; officer enters a payment; system updates the status and produces a receipt.

Req. ID	Requirement	Priority
HMS-F-040	The system shall generate an itemised bill for a visit (consultation fee plus services) and compute subtotal, tax at the configured rate and total.	High
HMS-F-041	The system shall record payments (cash, card or UPI), update the bill status to Unpaid, Partially Paid or Paid, and reject a payment greater than the outstanding balance.	High
HMS-F-042	The system shall generate a printable receipt with a unique receipt number for each payment.	Medium

FG6. Reports and Administration

Description and priority: Gives management summary reports and lets the Administrator manage users and review the audit trail. Priority: Medium.

Stimulus / response sequence: Administrator selects a report and date range, or a user action; system produces the result or updates the account.

Req. ID	Requirement	Priority
HMS-F-050	The system shall generate a daily appointment report and a revenue report for a chosen date range and export each as PDF or CSV.	Medium
HMS-F-060	The system shall let an Administrator create, deactivate and assign roles to user accounts; a deactivated user shall not be able to log in.	High
HMS-F-061	The system shall let an Administrator view the audit log filtered by user, date range and action type.	Medium

3.3 Non-Functional Requirements

3.3.1 Performance, Usability, Reliability and other Quality Attributes

Req. ID	Category	Requirement	Priority
HMS-NF-001	Performance	Patient search shall return results within 2 seconds for at least 95% of requests with 100,000 patient records and 100 concurrent users.	Medium
HMS-NF-002	Performance	Appointment booking (HMS-F-020)	Medium

Req. ID	Category	Requirement	Priority
		shall complete within 3 seconds for at least 95% of requests under the same load.	
HMS-NF-003	Capacity	The system shall support 200 concurrent user sessions for 30 minutes with an error rate below 1%.	Medium
HMS-NF-004	Availability	The system shall be available at least 99.5% of the time between 06:00 and 22:00 in any calendar month.	High
HMS-NF-005	Usability	A trained receptionist shall register a new patient in 3 minutes or less.	Medium
HMS-NF-006	Reliability	The system shall take an automated encrypted backup every 24 hours and shall be restorable within 4 hours, with at most 24 hours of data loss.	High
HMS-NF-007	Portability	The client shall run on Windows 10/11 and Ubuntu 22.04 and shall build with GCC 11+	Low

Req. ID	Category	Requirement	Priority
		and MSVC 2019+ using C++17.	
HMS-NF-008	Maintainability	Every service module shall have at least 70% line coverage from automated unit tests.	Medium
HMS-NF-009	Usability	In a test with 5 trained users, at least 90% of the core tasks (registering a patient and booking an appointment) shall be completed without help.	Medium

Relative preference of quality attributes: security and correctness of patient data first, then availability, then performance, then ease of use over ease of learning (staff are trained).

3.3.2 Safety Requirements

Wrong or missing clinical data can harm patients. The system therefore warns about recorded allergies (HMS-F-032), never deletes a visit record (records can only be corrected by adding a new entry that keeps the old one), and shows the patient name, Patient ID and date of birth on every clinical screen so that the doctor can confirm the right patient.

3.3.3 Security Requirements

Security objectives

ID	Objective	Description
SO-1	Confidentiality	Protect patient health information so that only authenticated users with a legitimate role can read it.
SO-2	Integrity	Ensure clinical, billing and patient data cannot be created, changed or deleted without

ID	Objective	Description
		authorisation, and that changes are detectable.
SO-3	Accountability	Make every access and change traceable to an individual user (non-repudiation).
SO-4	Availability and resilience	Keep the system usable and recoverable under abuse such as brute-force login attempts, session hijacking or injection attacks.

Security requirements

Req. ID	Requirement	Objective(s)	Priority
HMS-SEC-001	Passwords shall be stored only as salted hashes (Argon2id or bcrypt) and shall have at least 8 characters including an upper-case letter, a lower-case letter, a digit and a special character.	SO-1, SO-4	High
HMS-SEC-002	All communication between client and server shall use TLS 1.2 or higher; plain HTTP requests shall be refused.	SO-1, SO-2	High
HMS-SEC-003	A session shall expire after 10 minutes of inactivity, and a session token	SO-1, SO-4	High

Req. ID	Requirement	Objective(s)	Priority
	shall be invalid immediately after logout.		
HMS-SEC-004	All input shall be validated on the server, and all database access shall use parameterised queries, so that SQL injection and script injection attempts are rejected.	SO-2, SO-4	High
HMS-SEC-005	The system shall write an audit record (user, role, action, record ID, timestamp, source IP) for every login, record view, creation and update.	SO-3	High
HMS-SEC-006	Patient identifying and clinical fields shall be encrypted at rest with AES-256.	SO-1	High
HMS-SEC-007	Authorisation shall be checked on the server for every API call; hiding a function in the UI shall not be the only protection.	SO-1, SO-2	High
HMS-SEC-008	Audit records shall not contain passwords or	SO-1, SO-3	High

Req. ID	Requirement	Objective(s)	Priority
	clinical content and shall be readable only by the Administrator.		

Compliance: handling of PHI shall follow the DPDP Act 2023 and OWASP ASVS Level 2. Security validation of these requirements is in Section 5.1 of the Test Plan.

3.3.4 Business Rules and Domain Requirements

- BR-1: Only a Doctor may create clinical notes and prescriptions, and only a Doctor or the Patient concerned may view them; Receptionists and Billing Officers never see diagnoses.
- BR-2: A Patient may see only own appointments, records and bills.
- BR-3: Only an Administrator may create users, change roles or read the audit log.
- BR-4: Consultation slots are 30 minutes; a doctor cannot have two confirmed appointments in the same slot.
- BR-5: A bill, once fully paid, cannot be edited; corrections are made by a credit note.
- BR-6 (domain): Patient IDs and receipt numbers are never reused, even if a record is deactivated.

3.4 Design Constraints

- Language and platform: C++17; Qt 6 for the client; PostgreSQL for storage; OpenSSL for cryptography.
- Layered architecture with separate presentation, service and data layers so that each layer can be unit-tested.
- Coding standard: C++ Core Guidelines checked with clang-tidy; every module must be unit-testable.
- No patient data may be stored on the client machine after logout.

Appendix A: Glossary

Term	Definition
Appointment	A confirmed reservation of one 30-minute slot of a doctor for a patient.
Visit	The consultation that takes place for an appointment, holding symptoms, diagnosis and prescriptions.
Patient ID	Unique identifier PID-nnnnnnnn assigned at registration.
Audit log	Append-only record of who did what and when.

Term	Definition
Role	A set of permissions: Patient, Receptionist, Doctor, Billing Officer or Administrator.
Override reason	Text entered by a doctor to justify prescribing a medicine despite an allergy warning.

Appendix B: Field Layouts

Sample field layout for patient registration (HMS-F-010).

Field	Length	Data type	Description	Mandatory
Patient ID	12	Alphanumeric	System generated, format PID-nnnnnnnn	Y (auto)
Full name	60	String	Name of patient	Y
Date of birth	8	Date	DD-MM-YYYY, not in the future	Y
Gender	10	String	Female / Male / Other	Y
Phone	10	Numeric	Exactly 10 digits	Y
Address	120	String	Postal address	Y
Known allergies	200	String	Comma separated list	N
Emergency contact	60	String	Name and phone of contact	N

Sample report requirements: fields to include in each report.

Report	Fields
Daily Appointment Report	Appointment ID, Patient ID, Patient name, Doctor, Date, Slot, Status

Report	Fields
Revenue Report	Receipt number, Bill ID, Patient ID, Payment date, Payment mode, Amount, Tax, Total for range

Appendix C: Requirement Traceability Matrix

Architecture references are the component names in the Software Architecture and Design Specification (SAD). Code file references will be filled in during implementation (Part-2) and are currently TBD.

Sl. No	Req. ID	Brief description	Architecture ref.	Design ref.	Test case ID
1	HMS-F-001	Authenticate every user with a unique username and password	Auth Service (SAD 3.4)	SAD 4.2 Seq-1, 4.3 /auth	TC-AUTH-01
2	HMS-F-002	Lock account for 15 minutes after 5 failed attempts	Auth Service (SAD 3.4)	SAD 4.2 Seq-1, 4.3 /auth	TC-AUTH-03
3	HMS-F-003	Enforce role-based access (role-permission matrix)	Auth Service, API Controller (SAD 3.4)	SAD 4.2 Seq-1, 4.3 /auth	TC-RBAC-01
4	HMS-F-004	Generic message on invalid credentials	Auth Service (SAD 3.4)	SAD 4.2 Seq-1, 4.3 /auth	TC-AUTH-02
5	HMS-F-010	Register a new patient	Patient Service (SAD 3.4)	SAD 4.3 /patients	TC-PAT-01

Sl. No	Req. ID	Brief description	Architecture ref.	Design ref.	Test case ID
6	HMS-F-011	Reject invalid registration, name the offending field	Patient Service (SAD 3.4)	SAD 4.3 /patients	TC-PAT-02
7	HMS-F-012	Warn on duplicate patient before saving	Patient Service (SAD 3.4)	SAD 4.3 /patients	TC-PAT-03
8	HMS-F-013	Search patients by ID, name or phone	Patient Service (SAD 3.4)	SAD 4.3 /patients	TC-PAT-04
9	HMS-F-014	Update demographics, retain previous values	Patient Service, Audit Logger (SAD 3.4)	SAD 4.3 /patients	TC-PAT-05
10	HMS-F-020	Book an appointment	Appointment Service (SAD 3.4)	SAD 4.2 Seq-2, 4.3 /appointments	TC-APT-01
11	HMS-F-021	Prevent double-booking, show next 3 slots	Appointment Service (SAD 3.4)	SAD 4.2 Seq-2, 4.3 /appointments	TC-APT-02
12	HMS-F-022	Reject past and more-than-60-day bookings	Appointment Service (SAD 3.4)	SAD 4.2 Seq-2, 4.3 /appointments	TC-APT-03
13	HMS-F-023	Reschedule or cancel up to 2	Appointment	SAD 4.2 Seq-2, 4.3	TC-APT-04

Sl. No	Req. ID	Brief description hours before start	Architecture ref. Service (SAD 3.4)	Design ref. /appointments	Test case ID
14	HMS-F-024	Show doctor's daily appointment list	Appointment Service (SAD 3.4)	SAD 4.2 Seq-2, 4.3 /appointments	TC-APT-05
15	HMS-F-030	Record symptoms, diagnosis and notes for a visit	Clinical Service (SAD 3.4)	SAD 4.2 Seq-3, 4.3 /visits	TC-CLN-01
16	HMS-F-031	Add prescription items to a visit	Clinical Service (SAD 3.4)	SAD 4.2 Seq-3, 4.3 /visits	TC-CLN-02
17	HMS-F-032	Allergy warning with override reason	Clinical Service (SAD 3.4)	SAD 4.2 Seq-3, 4.3 /visits	TC-CLN-03
18	HMS-F-033	Display patient history in reverse chronological order	Clinical Service (SAD 3.4)	SAD 4.2 Seq-3, 4.3 /visits	TC-CLN-04
19	HMS-F-034	Patient views own records and bills	Clinical Service, Billing Service, API Controller (SAD 3.4)	SAD 4.3 /patients/me	TC-CLN-05
20	HMS-F-040	Generate itemised bill	Billing Service (SAD 3.4)	SAD 4.3 /bills	TC-BIL-01

Sl. No	Req. ID	Brief description	Architecture ref.	Design ref.	Test case ID
21	HMS-F-041	Record payments, update bill status	Billing Service (SAD 3.4)	SAD 4.3 /bills	TC-BIL-02
22	HMS-F-042	Generate printable receipt	Billing Service (SAD 3.4)	SAD 4.3 /bills	TC-BIL-03
23	HMS-F-050	Daily appointment and revenue reports	Report Service (SAD 3.4)	SAD 4.3 /reports	TC-RPT-01
24	HMS-F-060	Create, deactivate and assign roles to users	Auth Service (SAD 3.4)	SAD 4.2 Seq-1, 4.3 /auth	TC-ADM-01
25	HMS-F-061	View audit log with filters	Audit Logger (SAD 3.4)	SAD 4.4 Logging	TC-ADM-02
26	HMS-NF-001	Patient search within 2 seconds for 95% of requests	Data Access Layer, Patient Service	SAD 3.1, 4.1	TC-PERF-01
27	HMS-NF-002	Appointment booking within 3 seconds for 95% of requests	Appointment Service	SAD 3.1, 4.1	TC-PERF-02
28	HMS-NF-003	200 concurrent sessions for 30 minutes	API Controller, Data Access Layer	SAD 3.1, 4.1	TC-PERF-03

Sl. No	Req. ID	Brief description	Architecture ref.	Design ref.	Test case ID
29	HMS-NF-004	99.5% availability, 06:00 to 22:00	Backup Job	SAD 3.1, 4.1	TC-AVL-01
30	HMS-NF-005	Receptionist registers a patient in 3 minutes or less	Desktop UI Client	SAD 3.1, 4.1	TC-USA-01
31	HMS-NF-006	Automated encrypted backup every 24 hours	Backup Job	SAD 3.1, 4.1	TC-REL-01
32	HMS-NF-007	Client runs on Windows 10/11 and Ubuntu 22.04	Desktop UI Client	SAD 3.1, 4.1	TC-PORT-01
33	HMS-NF-008	At least 70% unit-test line coverage per service module	All Services	SAD 3.1, 4.1	TC-MNT-01
34	HMS-NF-009	90% of core tasks completed without help (5 users)	Desktop UI Client	SAD 3.1, 4.1, 4.5	TC-USA-01
35	HMS-SEC-001	Salted password hashes, password	Auth Service (SAD 3.9)	SAD 3.9, 4.4	TC-SEC-01

Sl. No	Req. ID	Brief description strength rules	Architecture ref.	Design ref.	Test case ID
36	HMS-SEC-002	TLS 1.2+ for all communication	API Controller (SAD 3.9)	SAD 3.9, 4.4	TC-SEC-02
37	HMS-SEC-003	10-minute session timeout, token invalid after logout	Auth Service (SAD 3.9)	SAD 3.9, 4.4	TC-SEC-03
38	HMS-SEC-004	Server-side validation, parameterised queries	API Controller, Data Access Layer (SAD 3.9)	SAD 3.9, 4.4	TC-SEC-04
39	HMS-SEC-005	Audit record for every login, view, creation and update	Audit Logger (SAD 3.9)	SAD 3.9, 4.4	TC-SEC-05
40	HMS-SEC-006	AES-256 encryption at rest for patient fields	Data Access Layer (SAD 3.9)	SAD 3.9, 4.4	TC-SEC-06
41	HMS-SEC-007	Server-side authorisation on every API call	API Controller, Auth Service (SAD 3.9)	SAD 3.9, 4.4	TC-SEC-07

Sl. No	Req. ID	Brief description	Architecture ref.	Design ref.	Test case ID
42	HMS-SEC-008	No passwords or clinical content in audit records; Administrator-only read	Audit Logger (SAD 3.9)	SAD 3.9, 4.4	TC-SEC-05